

Yueran (Hannah) Sun

Master's Student in Data Science · University of Washington

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EDUCATION

University of Washington, Seattle, WA.

M.S., Data Science

Sep 2025 - Mar 2027 (Expected)

GPA: 4.0/4.0

University of Michigan, Ann Arbor, MI.

B.S., Data Science · Minor: Mathematics · *University Honors*

Aug 2021 - May 2025

GPA: 3.71/4.0

RESEARCH EXPERIENCE

University of Washington - Language Accessibility Research Lab

Sep 2025–Present

Research Collaborator | Primary Collaborator: Anukriti Kumar | PI: Prof. Lucy Lu Wang

Seattle, WA

Productive Friction for Scientific Reading

- Designing a mixed-methods evaluation framework to investigate how AI-assisted productive-friction interfaces influence scientific paper comprehension and long-term retention.
- Reviewed prior HCI reading systems to develop the experimental design, hypotheses, and mixed-effects analysis plan for a three-condition longitudinal within-subject study.
- Developing immediate and delayed comprehension assessments with rubric-based scoring to evaluate surface, textbase, and situation-model learning outcomes across reading conditions.

AI-Assisted Plain-Language Adaptation

- Implemented automated segmentation and semantic alignment pipelines using LASER-3 embeddings and Vecalign, supporting comparative analysis of original and plain-language document pairs sourced from public-sector and nonprofit organizations.

University of Washington - Lab for Computing Cultural Heritage

Sep 2025–Present

Research Collaborator | PI: Prof. Benjamin Charles Germain Lee

Seattle, WA

- Developed an unsupervised computer vision pipeline using CLIP embeddings and DBSCAN to identify clusters of reprinted photographs across 1.5 million historical newspaper images from the Newspaper Navigator dataset.
- Built an API-driven image retrieval and validation workflow to surface clustered photographs on demand, enabling interactive inspection without downloading the full image dataset.
- Applied zero-shot CLIP classification to quantify semantic distributions across image clusters.

Humanities Data Science Summer Institute (American Institute of Physics Project)

Jun–Jul 2026

Graduate Research Assistant | PI: Prof. Benjamin Charles Germain Lee

Seattle, WA

- Mentored a team of four undergraduate researchers to improve discovery across archival collections held by the American Institute of Physics, including 1,700+ oral history interviews with physicists.
- Visualized relationships among physicists, laboratories, and institutions by transforming oral history transcripts into interactive knowledge graphs using GLiNER-based named entity recognition and relation extraction.
- Identified unnamed collaborators by connecting relationship-based references made by interviewees to external knowledge bases, strengthening attribution across historical scientific archives.

MobiDrop (Zhejiang) Co., Ltd.

Nov 2023–May 2024

Undergraduate Research Assistant | Mentor: Dr. Guantao Zheng

Remote

- Pretrained scGPT, a Transformer-based foundation model for single-cell gene expression modeling, on a curated dataset of 300,000 human blood cells from the CELLxGENE repository.
- Automated GPU training with Bash scripts for job submission, checkpointing, and experiment tracking.

PUBLICATIONS

- [C.1] Kumar, A., Glazko, K., **Sun, Y.**, Harniss, M., Wang, L. L., & Mankoff, J. (2026). *Beyond Readability Metrics: Plain Language Priorities in Disability Advocacy Organizations*. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*, 7864–7892.
- [S.1] Buccalon, B., **Sun, Y.**, & Lee, B. C. G. (under review). *Viral Images: Identifying Reprintings within 1.5 Million Photographs in Chronicling America*. Submitted to *Computational Humanities Research 2027 (CHR 2027)*.

INDUSTRY EXPERIENCE

Ternary

Jun–Aug 2025

Software Engineer Intern, Product Delivery Team | Supervisor: Kyle Rattet

Remote

- Integrated Meta Prophet into the cloud cost prediction API to complement the existing linear regression baseline with time-series forecasting.
- Engineered backend integrations between Go and Python services with schema validation, automated testing, and CI pipelines; deployed Terraform-managed staging environments on Google Cloud.
- Enhanced frontend client dashboards with interactive forecasting visualizations, including prediction intervals and breakdowns of the highest-cost services.

University of Michigan Multidisciplinary Design Program (ProQuest Team)

Jan–Dec 2024

Student Developer | Mentor: Prof. Sindhu Kutty | Industry Sponsors: John Dillon, Dan Hepp

Ann Arbor, MI

- Developed an automated document segmentation pipeline to extract article-level structure, including titles, bylines, body text, and reading order, improving indexing and search of historical newspapers.
- Built an MLOps pipeline integrating large language models, Gaussian mixture models, and computer vision for text classification, article boundary detection, and noise identification.
- Validated pipeline performance through A/B testing, reducing segmentation costs by 50% relative to manual workflows while increasing throughput to 12 pages per minute.

Pachira Information Technology (Hengqin) Co., Ltd.

May–Aug 2024

NLP Algorithm Intern | Supervisor: Dr. Chenglong Ma

Zhuhai, China

- Enhanced the RAG pipeline powering an in-car voice assistant for Toyota vehicles.
- Developed and deployed a warning-light query handling module through iterative testing in a vehicle simulator.
- Implemented multi-intent recognition and refusal logic to improve response accuracy and user safety.

TEACHING & INSTRUCTIONAL EXPERIENCE

University of Michigan - Department of Mathematics

Jan–May 2025

Grader, Matrix Algebra I | Supervisor: Prof. Ruiyuan Chen

Ann Arbor, MI

Developed grading rubrics and evaluated homework assignments.

University of Michigan - Math Learning Center

Sep–Dec 2023

Undergraduate Tutor | Supervisor: Irina Arakelian

Ann Arbor, MI

Tutored Calculus I, II, III, and Differential Equations.

LEADERSHIP & COMMUNITY ENGAGEMENT

Develop for Good

Remote

Program Lead

Mar 2026 - Present

Overseeing a portfolio of four interdisciplinary student volunteer teams, coordinating project execution, client communication, and mentor engagement across nonprofit software development initiatives.

Product Manager, PainUSA Team

Oct 2025–Feb 2026

Led five undergraduate student volunteers to prototype a web application for PainUSA, a nonprofit founded by Stanford University physicians, featuring an interactive clinician lookup map and chronic pain resources.

SKILLS

Languages: Python, R, SQL, Bash, C/C++, JavaScript/TypeScript, Go, HTML/CSS

Frameworks & Libraries: PyTorch, scikit-learn, Pandas, NumPy, OpenCV, XGBoost, LightGBM, LangChain, FastAPI, React, Node.js, Dask

Tools & Platforms: Linux, Git, Jupyter, Conda, Docker, GitHub Actions (CI/CD), Google Cloud Platform (GCP), Amazon Web Services (AWS), Terraform, MongoDB, IBM Db2, DBeaver, Tableau

Relevant Coursework: Machine Learning, Natural Language Processing, Data Mining, Database Management, Software Design, Web Systems, Data Visualization, User Interface Development, Applied Statistics and Experimental Design, Probability, Computation Theory, Discrete Mathematics, Matrix Algebra, Multivariable Calculus, Differential Equations, Financial Mathematics